

Constructor in Java – Theory Points

1. Definition:

A **constructor** is a special method used to initialize objects in Java. It is called when an object is created.

2. Name Rule:

A constructor **must have the same name** as the class and **does not have a return type**, not even `void`.

3. Types of Constructors:

- **Default Constructor:** Provided by Java if no constructor is defined. It has no parameters.
- **No-argument (Non-Parameterized) Constructor:** A constructor with no parameters, either default or user-defined.
- **Parameterized Constructor:** A constructor that takes arguments to initialize an object with specific values.
- **Copy Constructor:** Can be manually created to copy values from another object.

4. Constructor Overloading:

You can define **multiple constructors** in a class with different parameter lists (just like method overloading).

5. Automatic Invocation:

A constructor is **automatically called** when an object is created using the `new` keyword.

6. Usage:

Mainly used to **set initial values** for object attributes at the time of object creation.

7. this Keyword:

Often used in constructors to refer to the **current object** and resolve naming conflicts.

8. Constructor vs Method:

- No return type vs has return type.
- Automatically called vs manually called.
- Same name as class vs any valid name.

9. Constructors do NOT initialize static members in Java.

10. You can't call a constructor directly using its name

- You must use `new` to create an object. (Constructor gets called here)

10. If no constructor is defined by the user, the compiler automatically provides a default constructor.